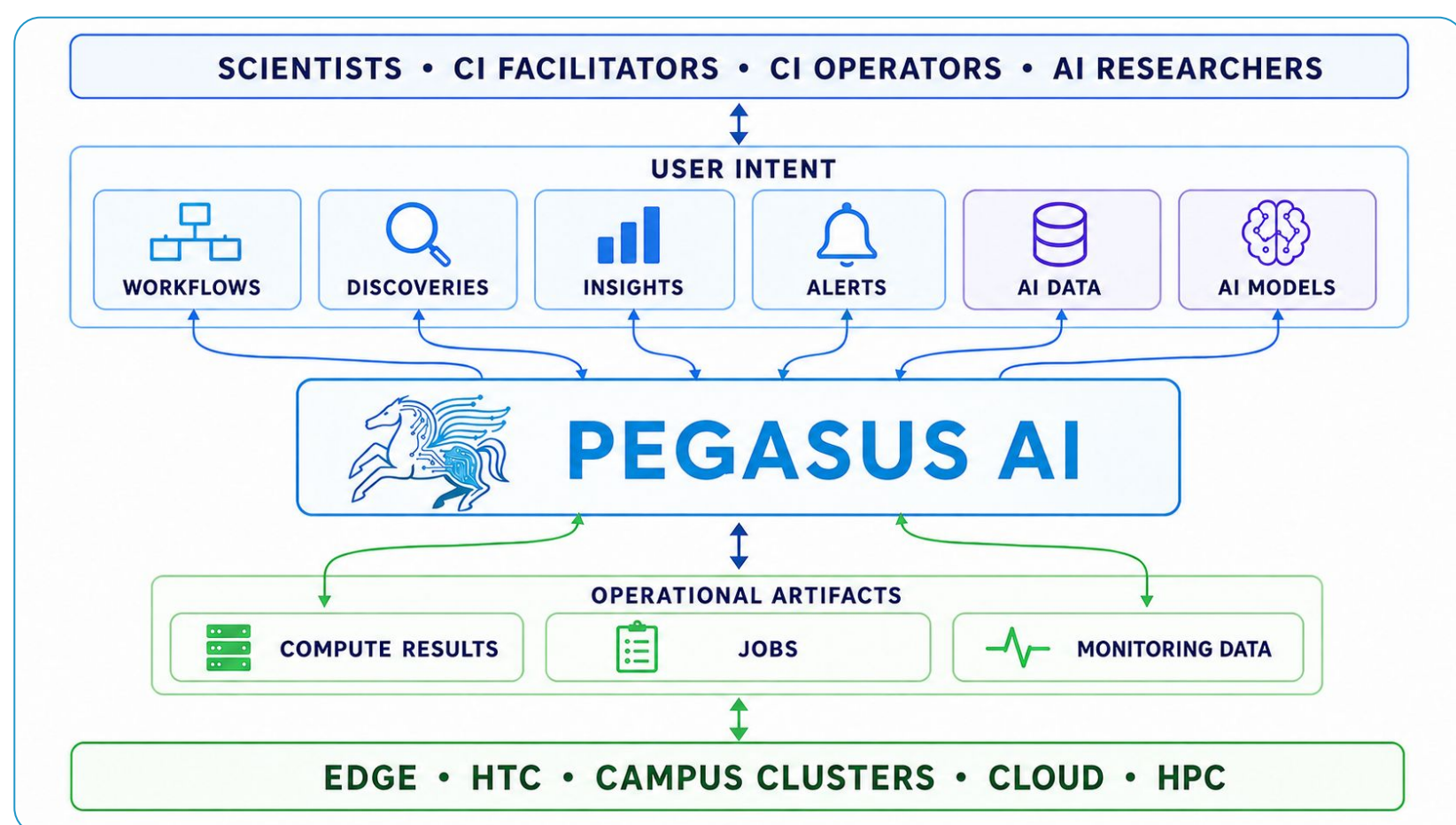
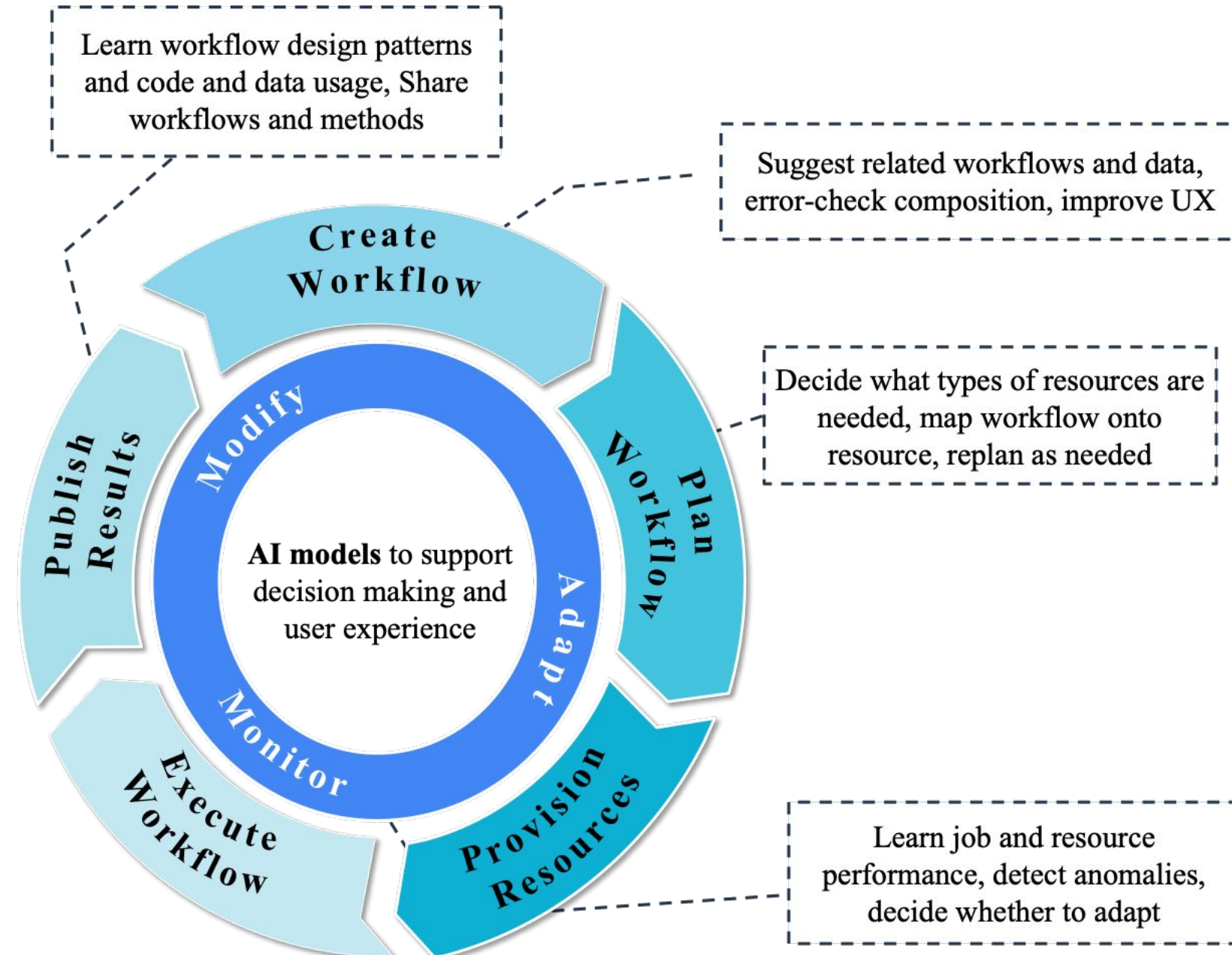


Introduction

Pegasus AI is developing an AI-driven scientific workflow management system that combines machine learning, large language models, online monitoring, and human guidance to support the workflow lifecycle. It helps scientists **create workflows, predict resource requirements, monitor execution, diagnose failures, and adapt execution plans**, making complex scientific workflows more efficient, resilient, and easier to manage



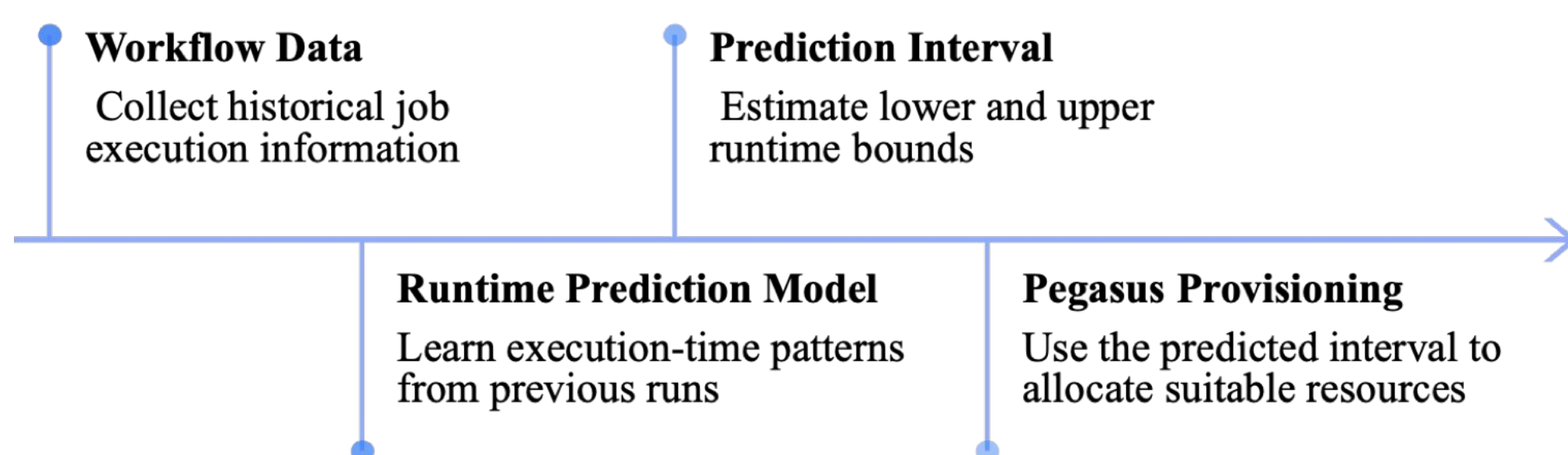
PegasusAI in the Workflow Lifecycle



Pegasus Runtime Prediction

Workflow jobs often have uncertain execution times, making resource provisioning and scheduling difficult. This project integrates a machine-learning framework into the Pegasus workflow lifecycle to predict a runtime interval for each job. The interval captures uncertainty and enables Pegasus to make more reliable resource-provisioning decisions

- ☐ Predicts lower and upper runtime bounds for workflow jobs
- ☐ Integrates predictions into the Pegasus planning lifecycle
- ☐ Supports uncertainty-aware resource provisioning
- ☐ Learns from historical workflow execution data
- ☐ Evaluates effects on resource utilization and workflow completion time



Pegasus Studio*

Pegasus AI Studio is an integrated environment that supports scientists across the workflow lifecycle, from defining research goals and generating Pegasus workflows with AI assistance to submitting them across computing resources, monitoring execution, diagnosing failures, and recommending corrective actions

01

Conversational AI Assistant

Supports workflow creation, answers Pegasus questions, explains execution results, and assists with failure diagnosis

03

Interactive Notebooks

Offers a browser-based environment for developing code, analyzing data, and accessing workflow inputs and outputs

02

Live Monitoring Dashboard

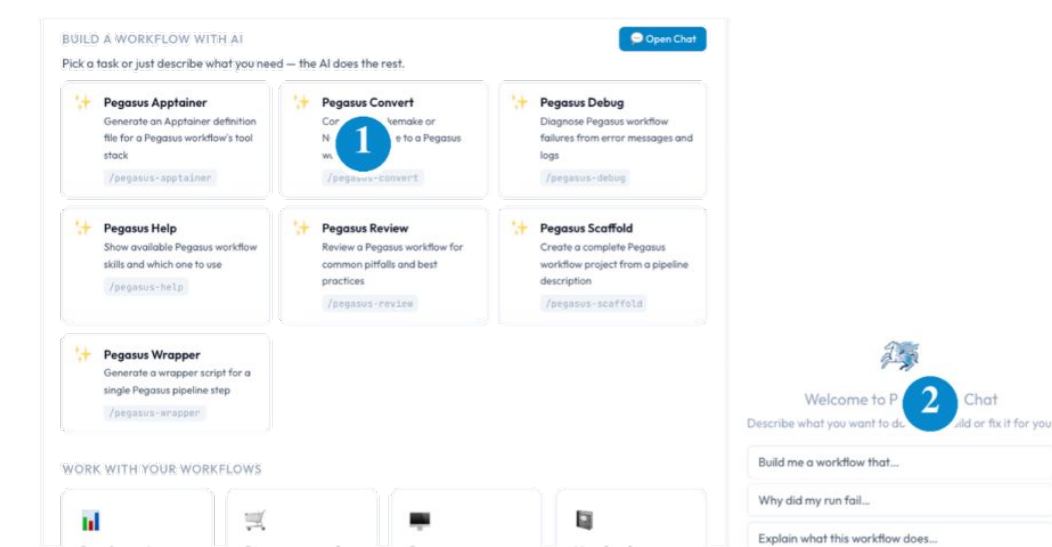
Provides workflow visualization, real-time job status, execution logs, errors, and alerts

04

Distributed Computing Access

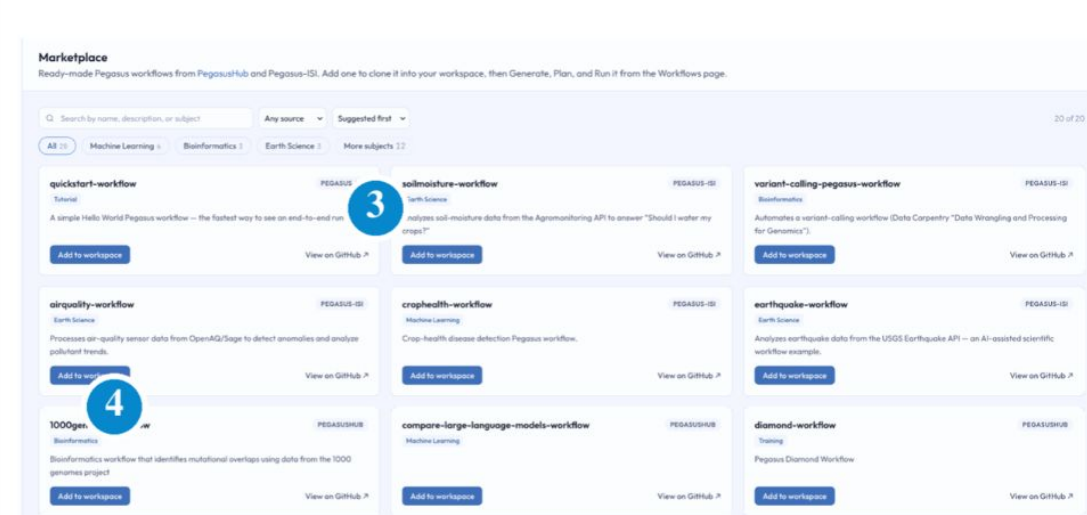
Enables workflows to execute across local, institutional, and shared computing resources

CREATE WITH AI



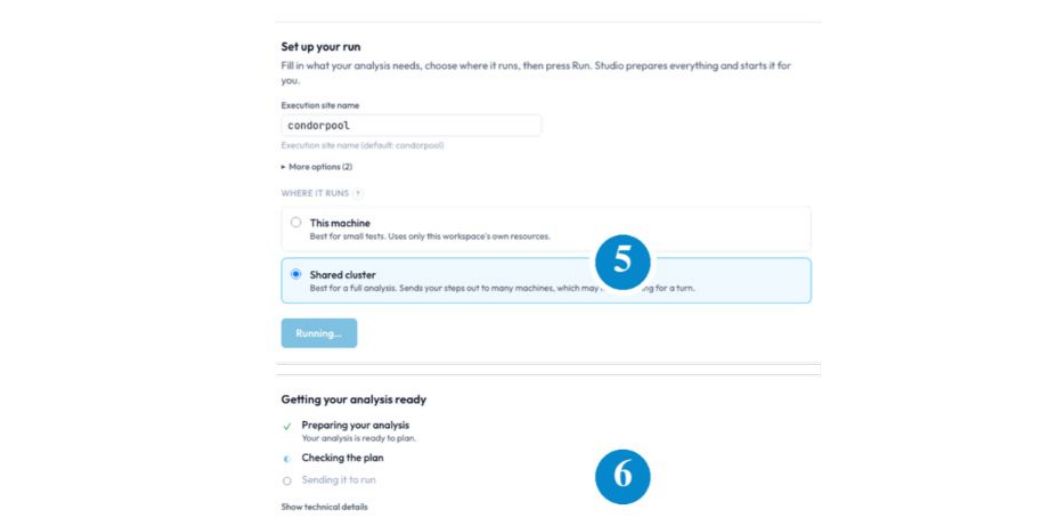
- 1 AI workflow skills: Build, convert, review, and debug.
- 2 PegasusAI Chat: Get guidance directly inside Studio.

DISCOVER AND REUSE



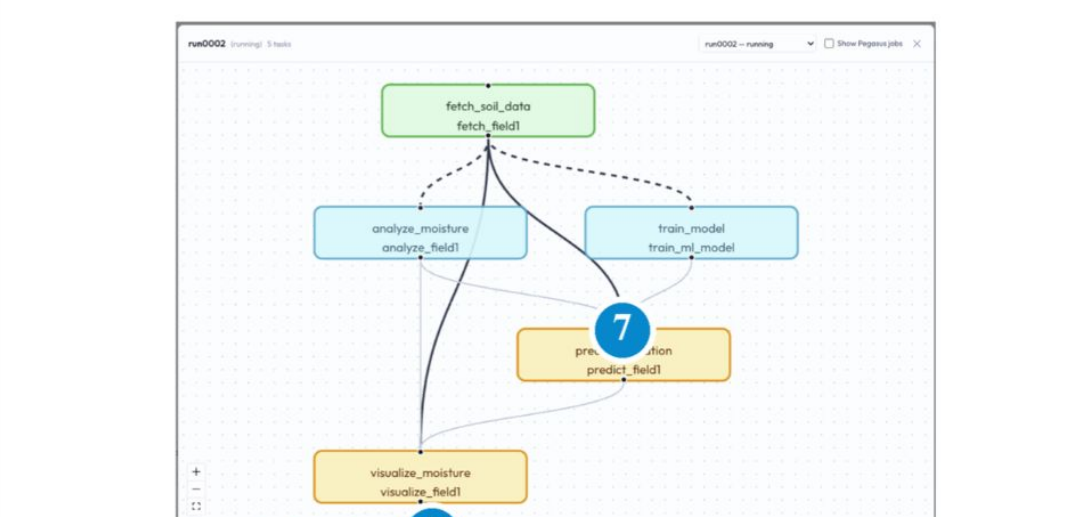
- 3 Workflow marketplace: Find reusable workflows by domain.
- 4 Add to workspace: Clone an example and start quickly.

CONFIGURE AND EXECUTE



- 5 Execution resources: Run locally or on a shared cluster.
- 6 Execution progress: Track planning and submission live.

MONITOR AND VISUALIZE



- 7 Running workflow DAG: Inspect active tasks and dependencies.
- 8 Live execution status: Track running, pending, and failed jobs.

* Pegasus studio : <https://studio.pegasus-ai.org>

PegasusAI Bot

PegasusAI Bot is a conversational AI assistant integrated into JupyterLab. It helps scientists understand and debug workflow code without leaving their notebook. The assistant analyzes the selected cell, its outputs, and execution errors to provide context-aware explanations and suggested corrective actions

- ☐ Explains workflow errors in accessible language
- ☐ Provides debugging guidance inside JupyterLab
- ☐ Suggests possible corrections and next steps
- ☐ Reduces manual searching across logs and documentation
- ☐ Collects optional feedback to improve future assistance
- ☐ Maintains human review before applying changes

